

WASP-35b

R-1060

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_251030 · created 2026-07-30T11:39:54+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460978.8547 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1281 +/- 0.0062
Transit depth (Rp/Rs)^2	1.64 +/- 0.16 [%]
Orbital Inclination (inc)	88.7 +/- 1.13
Airmass coefficient 1 (a1)	1.0378 +/- 0.005
Airmass coefficient 2 (a2)	-0.028 +/- 0.04
Scatter in the residuals of the lightcurve fit is	0.48 %
Best Comparison Star	#2 - [92, 422]
Optimal Aperture	3.84
Optimal Annulus	13.91
Transit Duration (day)	0.0556 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1281 $\pm$ 0.0062 vs 0.1241 (deviation 0.58 σ)
Duration fit vs published	0.0556 d vs 0.1317 d (deviation 14.87 σ)
Mid-time O-C	-11.3 min
Depth SNR	18.6
Residual scatter	0.48 %

Light curve

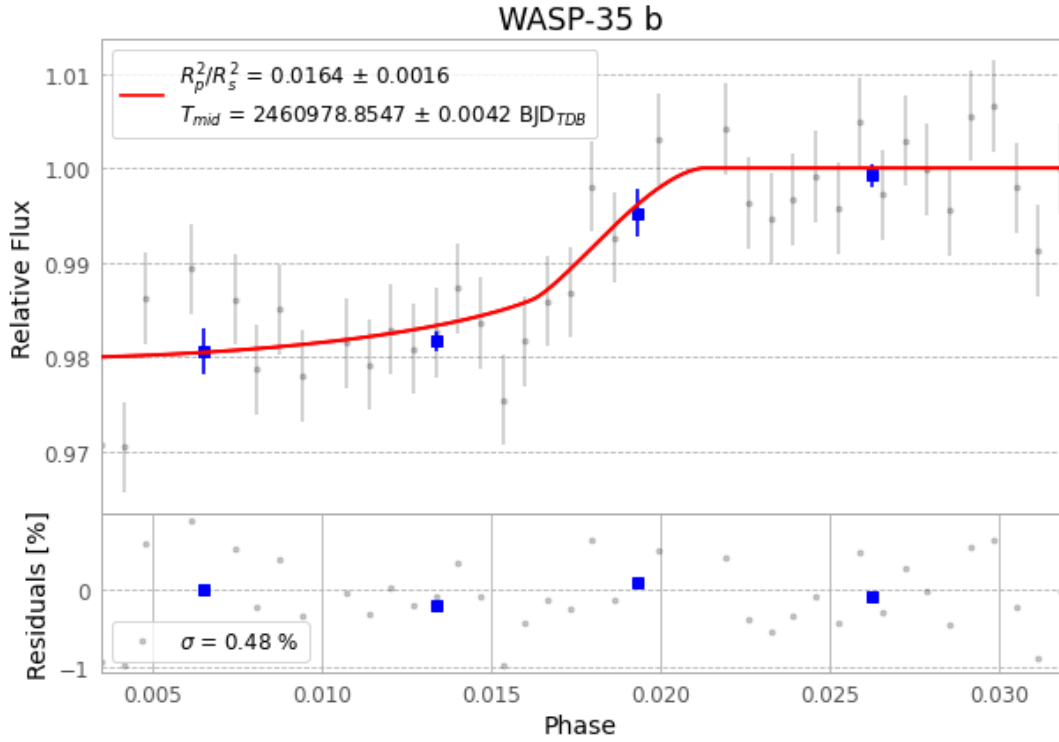


Figure 1 — FinalLightCurve_WASP-35 b_2025-10-30.png (SHA-256 be63eded15d001d6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 271], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a9430530aaae6fc4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

39 FITS frames (26 MB), WASP-35251030083915.FITS → WASP-35251030104815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-251030.log (c7097c4f57e0...), FinalLightCurve_WASP-35 b_2025-10-30.png (be63eded15d0...), FinalLightCurve_WASP-35 b_2025-10-30.csv (3a05f28e61d5...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 11:39Z.